

Solar Panel Degradation Predictor

Prediction Report

Result ID: #3
Generated: 2026-07-20 08:32:35 UTC

Predicted Solar Panel Efficiency

0.4909 (49.1%)

Maintenance Recommendation

No immediate maintenance needed - monitor periodically

Input Feature Values

Feature		Feature	
temperature	35.0000	module_temperature	40.0000
irradiance	600.0000	cloud_coverage	20.0000
humidity	60.0000	wind_speed	3.5000
panel_age	10.0000	pressure	1013.0000
maintenance_count	2.0000	string_id	A1
soiling_ratio	0.3000	error_code	E00
voltage	45.0000	installation_type	fixed
current	8.0000		

Model: Gradient Boosting Regressor (R2=0.790, RMSE=0.048, test set 4,000 samples). Primary driver: solar irradiance (feature importance 0.674).
Maintenance thresholds derived from EDA (dataset mean efficiency ~0.52).